

Solution Requirements

Date	15 March 2026
Team ID	UNM293SmartStudyAI
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

This document lists what ComicCraft will do and the quality standards it must meet. Functional requirements describe system features, while non-functional requirements describe how well and securely the system should operate.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can register, log in, and log out securely. The system validates credentials and provides password reset support.	High
2	Authorization levels	Registered users can create and manage their own comic projects. Administrators can manage users and review system activity.	Medium
3	External interfaces	Connect to Google Gemini models through an API to generate story ideas, dialogue, scene descriptions, and image prompts. Store project data in a database.	High
4	Transactions processing	Accept user prompts and preferences, send requests to Gemini, process responses, and save generated stories and comic drafts. Show clear errors if a request fails.	High
5	Reporting	Display project history and generation status. Allow users to view, copy, and export completed comic scripts or story drafts.	Medium
6	Business rules, etc.	Generate content according to the selected genre, characters, tone, and number of scenes. Keep character details consistent across scenes and allow users to edit or regenerate content.	High
7	Compliance to laws or regulations	Protect user account and project data. Follow applicable privacy, copyright, and AI-service usage requirements; do not present generated content as guaranteed original or legally cleared.	High
8	Other	Provide a simple interface for entering prompts, selecting story settings, previewing scenes, editing dialogue, and saving drafts.	Medium

Solution Requirements (Continued)

Step 2: Non-Functional Requirements (NFR)

These requirements define the performance, security, reliability, and usability expected from ComicCraft.

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The application should respond quickly to user actions and show progress while AI content is being generated.	Most pages load within 3 seconds; generation status appears within 2 seconds.
2	Scalability	The system should support growth in users, saved projects, and AI requests without major changes to its design.	Support at least 100 concurrent users in the initial deployment; allow capacity to be increased.
3	Security & Data Privacy	Use secure authentication, HTTPS communication, protected API credentials, and access controls. Avoid exposing private prompts or project data.	HTTPS for all network traffic; API keys stored server-side; users can access only their own projects.
4	Reliability & Availability	Handle temporary API or network failures gracefully and preserve saved work. Provide clear retry options.	Target 99% monthly availability, excluding planned maintenance and third-party outages.
5	Usability & Accessibility	Provide a clear, responsive interface that works on mobile and desktop. Use readable text, understandable labels, and keyboard-friendly controls.	Core tasks can be completed with clear navigation; responsive layout for common mobile and desktop screen sizes.
6	Other: Maintainability & Portability	Keep the application modular so the AI integration, interface, and data storage can be updated independently.	Document setup and key modules; use configuration for environment-specific settings; maintainable code structure.

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